

RLM Local Ollama Benchmark Guide And Results

Scope

This document covers only the local Ollama benchmark batch completed on qwen2.5:7b. It does not use or summarize any earlier Gemini runs.

Environment

- Workspace: /Users/abhigyansekhar/Desktop/RLM-FULL
- Backend: ollama through the OpenAI-compatible local endpoint
- Model: qwen2.5:7b
- Batch folder: /Users/abhigyansekhar/Desktop/RLM-FULL/benchmark_runs/20260403-121151-ollama-qwen2.5-7b
- Run plan: 5 paired LLM / RLM runs for each benchmark

Setup

```
cd /Users/abhigyansekhar/Desktop/RLM-FULL
source .venv/bin/activate
uv pip install -e .
open -a Ollama
/Applications/Ollama.app/Contents/Resources/ollama pull qwen2.5:7b
export BENCHMARK_BACKEND=ollama
export OLLAMA_BASE_URL=http://127.0.0.1:11434/v1
export OLLAMA_MODEL_NAME=qwen2.5:7b
export OLLAMA_API_KEY=ollama
export BENCHMARK_COOLDOWN_SECONDS=0
```

How The Run Artifacts Are Organized

The completed batch is already organized one folder per test. Each test folder contains the five LLM logs and five RLM logs as markdown files.

Example layout:

```
benchmark_runs/20260403-121151-ollama-qwen2.5-7b/
  got/
    run_001_llm.md
    run_001_rlm.md
    ...
  authproxy/
    run_001_llm.md
    run_001_rlm.md
    ...
  clinical/
    run_001_llm.md
```

```

run_001_rlm.md
...
launch_note_app/
run_001_llm.md
run_001_rlm.md
...
pdf_cersei_warning/
run_001_llm.md
run_001_rlm.md
...
tsp_branch_bound/
run_001_llm.md
run_001_rlm.md
...
stochastic_tsp/
run_001_llm.md
run_001_rlm.md
...
SUMMARY.md
aggregate_metrics.csv
raw_data.csv

```

Commands Used

Single pair example:

```

.venv/bin/python llm-test/test_tsp_llm_only.py
.venv/bin/python rlm-test/test_tsp_branch_bound.py

```

Full paired batch:

```

.venv/bin/python run_benchmark_pairs.py --backend ollama --model qwen2.5:7b --runs 5

```

Aggregate Results

			Avg Wall (s)	Median Wall (s)	Avg Input Tokens	Avg Output Tokens	Avg Total Tokens
Benchmark	Mode	Success					
Battle of the Bastards	LLM	5/5 (100%)	7.803	7.756	275.000	38.600	313.600
Battle of the Bastards	RLM	5/5 (100%)	108.379	73.517	9145.000	440.000	9585.000

Benchmark	Mode	Success	Avg Wall (s)	Median Wall (s)	Avg Input Tokens	Avg Output Tokens	Avg Total Tokens
AuthProxy	LLM	5/5 (100%)	23.689	23.109	1082.000	166.000	1248.000
Long Context							
AuthProxy	RLM	5/5 (100%)	179.107	173.042	11749.000	1251.600	13000.600
Long Context							
Clinical	LLM	5/5 (100%)	104.282	111.993	1645.000	1179.600	2824.600
Long Context							
Clinical	RLM	5/5 (100%)	137.302	52.487	11503.800	1017.200	12521.000
Long Context							
Launch Note	LLM	5/5 (100%)	91.281	81.166	257.000	1044.600	1301.600
App Launch Note							
App Launch Note	RLM	5/5 (100%)	273.112	276.742	10956.800	2154.200	13111.000
App PDF							
Cersei Warning	LLM	5/5 (100%)	132.849	133.707	4096.000	912.400	5008.400
PDF Cersei Warning							
PDF Cersei Warning	RLM	5/5 (100%)	212.396	191.413	19444.400	951.600	20396.000
Under-specified TSP							
Under-specified TSP	LLM	5/5 (100%)	108.994	106.786	209.000	1115.600	1324.600
Under-specified TSP							
Under-specified TSP	RLM	5/5 (100%)	510.680	369.738	20109.200	3406.000	23515.200
Stochastic Adaptive TSP							
Stochastic Adaptive TSP	LLM	5/5 (100%)	164.309	185.630	318.000	1052.600	1370.600

Benchmark	Mode	Success	Avg Wall (s)	Median Wall (s)	Avg Input Tokens	Avg Output Tokens	Avg Total Tokens
Stochastic Adaptive TSP	RLM	4/5 (80%)	843.710	700.832	24390.000	3300.500	27690.500

Wall Time Graph

Benchmark	Mode	Avg Wall (s)	Relative Bar
Battle of the Bastards	LLM	7.803	#
Battle of the Bastards	RLM	108.379	###
AuthProxy Long Context	LLM	23.689	#
AuthProxy Long Context	RLM	179.107	#####
Clinical Long Context	LLM	104.282	###
Clinical Long Context	RLM	137.302	####
Launch Note App	LLM	91.281	###
Launch Note App	RLM	273.112	#####
PDF Cersei Warning	LLM	132.849	###
PDF Cersei Warning	RLM	212.396	#####
Under-specified TSP	LLM	108.994	###
Under-specified TSP	RLM	510.680	#####
Stochastic Adaptive TSP	LLM	164.309	###
Stochastic Adaptive TSP	RLM	843.710	#####

Token Usage Graph

Benchmark	Mode	Avg Total Tokens	Relative Bar
Battle of the Bastards	LLM	313.600	#
Battle of the Bastards	RLM	9585.000	#####
AuthProxy Long Context	LLM	1248.000	#
AuthProxy Long Context	RLM	13000.600	#####
Clinical Long Context	LLM	2824.600	##
Clinical Long Context	RLM	12521.000	#####
Launch Note App	LLM	1301.600	#
Launch Note App	RLM	13111.000	#####
PDF Cersei Warning	LLM	5008.400	###
PDF Cersei Warning	RLM	20396.000	#####
Under-specified TSP	LLM	1324.600	#
Under-specified TSP	RLM	23515.200	#####

Benchmark	Mode	Avg Total Tokens	Relative Bar
Stochastic Adaptive TSP	LLM	1370.600	#
Stochastic Adaptive TSP	RLM	27690.500	#####

Raw Data Notes

- All values below come from the markdown logs inside the completed local batch folder.
- `stochastic_tsp` run 004 on the RLM side exited with `-9`, so it is included in the raw table but excluded from aggregate averages.

Raw Data Table

Benchmark	Run	Mode	Exit	Wall (s)	Exec (s)	Input	Output	Total	Log
Battle of the Bas-tards	001	LLM	0	9.832	9.832	275	56	331	got/run_001_llm.md
	001	RLM	0	73.517	73.048	8,816	299	9,115	got/run_001_rlm.md
Battle of the Bas-tards	002	LLM	0	7.135	7.135	275	41	316	got/run_002_llm.md
	002	RLM	0	244.427	197.253	13,982	936	14,918	got/run_002_rlm.md
Battle of the Bas-tards	003	LLM	0	7.756	7.756	275	33	308	got/run_003_llm.md

Benchmark	Run	Mode	Exit	Wall (s)	Exec (s)	Input	Output	Total	Log
Battle of the Bas-tards	003	RLM	0	73.046	72.665	8,753	367	9,120	got/run_003_rlm.md
Battle of the Bas-tards	004	LLM	0	5.573	5.573	275	33	308	got/run_004_llm.md
Battle of the Bas-tards	004	RLM	0	60.789	60.714	5,483	289	5,772	got/run_004_rlm.md
Battle of the Bas-tards	005	LLM	0	8.717	8.717	275	30	305	got/run_005_llm.md
Battle of the Bas-tards	005	RLM	0	90.114	89.877	8,691	309	9,000	got/run_005_rlm.md
AuthProxy Long Con-text	001	LLM	0	30.373	30.373	1,082	178	1,260	authproxy/run_001_llm.md
AuthProxy Long Con-text	001	RLM	0	169.596	169.158	6,619	1,414	8,033	authproxy/run_001_rlm.md
AuthProxy Long Con-text	002	LLM	0	20.656	20.656	1,082	145	1,227	authproxy/run_002_llm.md
AuthProxy Long Con-text	002	RLM	0	219.642	173.374	13,844	1,741	15,585	authproxy/run_002_rlm.md

Benchmark	Run	Mode	Exit	Wall (s)	Exec (s)	Input	Output	Total	Log
AuthProxy Long Con- text	003	LLM	0	20.201	20.201	1,082	149	1,231	authproxy/run_003_llm.md
AuthProxy Long Con- text	003	RLM	0	146.474	118.017	13,488	990	14,478	authproxy/run_003_rlm.md
AuthProxy Long Con- text	004	LLM	0	24.106	24.106	1,082	171	1,253	authproxy/run_004_llm.md
AuthProxy Long Con- text	004	RLM	0	173.042	112.295	13,385	937	14,322	authproxy/run_004_rlm.md
AuthProxy Long Con- text	005	LLM	0	23.109	23.109	1,082	187	1,269	authproxy/run_005_llm.md
AuthProxy Long Con- text	005	RLM	0	186.782	153.458	11,409	1,176	12,585	authproxy/run_005_rlm.md
Clinical Long Con- text	001	LLM	0	137.570	137.570	1,645	1,583	3,228	clinical/run_001_llm.md
Clinical Long Con- text	001	RLM	0	35.334	31.440	10,181	125	10,306	clinical/run_001_rlm.md
Clinical Long Con- text	002	LLM	0	111.993	111.993	1,645	1,312	2,957	clinical/run_002_llm.md
Clinical Long Con- text	002	RLM	0	52.487	44.638	10,392	351	10,743	clinical/run_002_rlm.md

Benchmark	Run	Mode	Exit	Wall (s)	Exec (s)	Input	Output	Total	Log
Clinical 003 Long Con- text	003	LLM	0	112.244	112.244	1,645	1,234	2,879	clinical/run_003_llm.md
Clinical 003 Long Con- text	003	RLM	0	34.254	32.456	10,160	123	10,283	clinical/run_003_rlm.md
Clinical 004 Long Con- text	004	LLM	0	101.873	101.873	1,645	1,270	2,915	clinical/run_004_llm.md
Clinical 004 Long Con- text	004	RLM	0	343.306	281.513	15,177	2,883	18,060	clinical/run_004_rlm.md
Clinical 005 Long Con- text	005	LLM	0	57.730	57.730	1,645	499	2,144	clinical/run_005_llm.md
Clinical 005 Long Con- text	005	RLM	0	221.131	220.670	11,609	1,604	13,213	clinical/run_005_rlm.md
Launch 001 Note App	001	LLM	0	81.166	81.166	257	978	1,235	launch_note_app/run_001_llm.md
Launch 001 Note App	001	RLM	0	226.218	147.923	11,223	2,144	13,367	launch_note_app/run_001_rlm.md
Launch 002 Note App	002	LLM	0	88.222	88.222	257	1,129	1,386	launch_note_app/run_002_llm.md
Launch 002 Note App	002	RLM	0	276.742	232.978	13,106	2,843	15,949	launch_note_app/run_002_rlm.md
Launch 003 Note App	003	LLM	0	58.346	58.346	257	879	1,136	launch_note_app/run_003_llm.md
Launch 003 Note App	003	RLM	0	102.077	101.825	5,808	1,416	7,224	launch_note_app/run_003_rlm.md

Benchmark	Run	Mode	Exit	Wall (s)	Exec (s)	Input	Output	Total	Log
Launch Note App	004	LLM	0	72.258	72.258	257	837	1,094	launch_note_app/run_004_llm.md
Launch Note App	004	RLM	0	464.365	304.048	14,161	2,426	16,587	launch_note_app/run_004_rlm.md
Launch Note App	005	LLM	0	156.415	156.415	257	1,400	1,657	launch_note_app/run_005_llm.md
Launch Note App	005	RLM	0	296.160	291.026	10,486	1,942	12,428	launch_note_app/run_005_rlm.md
PDF Cer-sei Warning	001	LLM	0	135.947	135.947	4,096	728	4,824	pdf_cersei_warning/run_001_llm
PDF Cer-sei Warning	001	RLM	0	191.413	191.159	10,247	903	11,150	pdf_cersei_warning/run_001_rlm
PDF Cer-sei Warning	002	LLM	0	116.163	116.163	4,096	692	4,788	pdf_cersei_warning/run_002_llm
PDF Cer-sei Warning	002	RLM	0	231.660	184.283	21,579	899	22,478	pdf_cersei_warning/run_002_rlm
PDF Cer-sei Warning	003	LLM	0	133.442	133.442	4,096	816	4,912	pdf_cersei_warning/run_003_llm
PDF Cer-sei Warning	003	RLM	0	318.367	277.481	22,295	1,289	23,584	pdf_cersei_warning/run_003_rlm

Benchmark	Run	Mode	Exit	Wall (s)	Exec (s)	Input	Output	Total	Log
PDF Cer-sei Warning	004	LLM	0	133.707	133.707	4,096	1,085	5,181	pdf_cersei_warning/run_004_llm.m
PDF Cer-sei Warning	004	RLM	0	139.048	116.216	21,496	777	22,273	pdf_cersei_warning/run_004_rlm.m
PDF Cer-sei Warning	005	LLM	0	144.988	144.988	4,096	1,241	5,337	pdf_cersei_warning/run_005_llm.m
PDF Cer-sei Warning	005	RLM	0	181.494	123.025	21,605	890	22,495	pdf_cersei_warning/run_005_rlm.m
Under-specified TSP	001	LLM	0	83.021	83.021	209	1,154	1,363	tsp_branch_bound/run_001_llm.m
Under-specified TSP	001	RLM	0	253.519	253.280	12,514	2,661	15,175	tsp_branch_bound/run_001_rlm.m
Under-specified TSP	002	LLM	0	85.607	85.607	209	1,055	1,264	tsp_branch_bound/run_002_llm.m
Under-specified TSP	002	RLM	0	369.738	296.882	23,510	2,468	25,978	tsp_branch_bound/run_002_rlm.m
Under-specified TSP	003	LLM	0	127.640	127.640	209	1,057	1,266	tsp_branch_bound/run_003_llm.m
Under-specified TSP	003	RLM	0	705.174	697.629	22,245	4,097	26,342	tsp_branch_bound/run_003_rlm.m
Under-specified TSP	004	LLM	0	141.916	141.916	209	1,103	1,312	tsp_branch_bound/run_004_llm.m

Benchmark	Run	Mode	Exit	Wall (s)	Exec (s)	Input	Output	Total	Log
Under-specified TSP	004	RLM	0	970.961	846.544	20,689	5,498	26,187	tsp_branch_bound/run_004_rlm.md
Under-specified TSP	005	LLM	0	106.786	106.786	209	1,209	1,418	tsp_branch_bound/run_005_llm.md
Under-specified TSP	005	RLM	0	254.008	192.689	21,588	2,306	23,894	tsp_branch_bound/run_005_rlm.md
Stochastic Adaptive TSP	001	LLM	0	98.209	98.209	318	1,059	1,377	stochastic_tsp/run_001_llm.md
Stochastic Adaptive TSP	001	RLM	0	633.152	582.324	24,446	4,417	28,863	stochastic_tsp/run_001_rlm.md
Stochastic Adaptive TSP	002	LLM	0	106.263	106.263	318	999	1,317	stochastic_tsp/run_002_llm.md
Stochastic Adaptive TSP	002	RLM	0	1342.957	1238.970	24,535	3,297	27,832	stochastic_tsp/run_002_rlm.md
Stochastic Adaptive TSP	003	LLM	0	224.292	224.292	318	1,170	1,488	stochastic_tsp/run_003_llm.md
Stochastic Adaptive TSP	003	RLM	0	768.512	746.877	23,597	2,772	26,369	stochastic_tsp/run_003_rlm.md
Stochastic Adaptive TSP	004	LLM	0	207.149	207.149	318	1,061	1,379	stochastic_tsp/run_004_llm.md
Stochastic Adaptive TSP	004	RLM	-9	-	-	-	-	-	stochastic_tsp/run_004_rlm.md

Benchmark	Run	Mode	Exit	Wall (s)	Exec (s)	Input	Output	Total	Log
Stochastic Adap- tive TSP	005	LLM	0	185.630	185.630	318	974	1,292	stochastic_tsp/run_005_llm.md
Stochastic Adap- tive TSP	005	RLM	0	630.217	589.147	24,982	2,716	27,698	stochastic_tsp/run_005_rlm.md

Output Files

- Raw CSV: `raw_data.csv`
- Aggregate CSV: `aggregate_metrics.csv`
- Batch summary: `SUMMARY.md`